Mouhammed Soumaille

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Mathematics and computer engineering undergraduate with research experience in probabilistic modeling and empirical evaluation in Python. Interested in empirical AI safety, especially rigorous evaluation of model behavior and interpretability, where probability and statistics can make results more trustworthy.

Education

American University of Beirut (AUB) Expected Spring 2029
B.S. in Computer Science and Engineering & Pure Mathematics *Beirut, Lebanon*

- **Relevant coursework:** Machine Learning; Graduate Probability Theory; Graduate Convex Optimization; Theoretical Foundations of Security & Privacy; Complexity Theory; Data Structures & Algorithms; Real Analysis; Advanced Linear Algebra

Research Experience

Research Assistant — Heavy-Tailed Financial Risk Fall 2025 – Spring 2026
American University of Beirut *Advisors: Prof. Ibrahim Abou Faycal, Prof. Jihad Fahs*

- Developed the **Extreme Loss Quantile Bound (ELQB)**, a spectral risk measure for extreme losses under heavy-tailed distributions; proved coherence, normalization, finiteness, and limiting behavior.
- Extended the framework to portfolio-level and conditional settings, introducing **Co-ELQB** and a systemic amplification ratio that quantifies how much an asset's tail risk grows under system-wide stress.
- Built the empirical pipeline in **Python** and evaluated it on five large-cap U.S. equities (2020–2025), including a rolling-window analysis of the COVID-19 crash; documented results in a report and slides.

Research Assistant — Quantum Error Correction Fall 2026 – Present
American University of Beirut *Advisor: Prof. Wael Itani*

- Computing entanglement polytopes of multipartite quantum systems from reduced density matrices and local spectra, using group-theoretic techniques; studying their relevance to quantum error-correcting codes.

Projects

Do Linear Truth Probes Generalize? — LLM Interpretability Oct 2026
Independent project *github.com/mouhammed-soumaille/truth-probes*

- Trained linear **truth probes** on every layer of Qwen2.5-0.5B and 1.5B (Geometry of Truth datasets) and tested how they transfer across datasets, with bootstrap intervals and ranges over 5 random splits.
- Replicated the negation failure: probes that are near-perfect in-distribution are confidently wrong on negated statements; a pair test showed they give a statement and its negation the same answer 96–100% of the time.
- Showed that four layers **tied** on validation accuracy (0.997) give 0.05–0.53 accuracy on negated statements, and traced a reproducibility bug to unstable layer tie-breaking.

DAYFIT — Visual Clothing Search Fall 2026
American University of Beirut *In progress*

- Building an image-retrieval system that returns the top-5 most similar garments from a 1,000–3,000 image catalog using pretrained **CLIP** embeddings.
- Designing the evaluation: comparing CLIP against a color-histogram baseline on Precision@5 and latency, and stress-testing **robustness** to shifts in lighting, background, and camera angle.

Technical Skills

Programming: Python (primary), C++, MATLAB
ML & scientific computing: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, SciPy, pandas, CVXPY
Tools: Git, Linux/Bash, Docker, FastAPI, Streamlit, LaTeX

Teaching & Communication

Tutor — Probability & Statistics; Analysis I Fall 2026
Mathematics Tutoring Center, AUB

- Lead review sessions on probabilistic reasoning and proof-based analysis for students at different levels.